## Multiple Choice:

1. In a track-and-field event, an athlete runs exactly once around an oval track, a total distance of 500 m. Find the runner's displacement for the race.

   A. 0
   B. 500 m
   C. equal to the perimeter of the track
   D. none of the above

2. In a track-and-field event, an athlete runs exactly once around an oval track, a total distance of 500 m. If the runner completes the race in 1 minute and 20 seconds, find the magnitude of the average velocity?

   A. 0 C. 5.2 m/s
   B. 6.3 m/s D. 9.1 m/s

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4. A rock is dropped off a cliff and strikes the ground with an impact velocity of 30 m/s. How high was the cliff?

   A. 20 m C. 80 m
   B. 45 m D. 32 m

5. What net force is required to maintain a 5,000 kg object moving at a constant velocity of magnitude 7,500 m/s?

   A. 0 C. 37.5 MN
   B. 20 MN D. 60 MN

6. An object feels two forces: one of magnitude 8 N pulling to the left and one of magnitude 20 N pulling to the right. If the object's mass is 4 kg, what is its acceleration?

   A. 0 C. 5 $m/s^2$
   B. 3 $m/s^2$ D. 8 $m/s^2$

7. A woman riding a bicycle collides head-on with a parked school bus. Which object feels greater force?

   A. The woman experiences greater force
   B. The bus experiences greater force
   C. The woman and bus experience the same force
   D. There is not enough information

8. An artificial satellite of mass m travels at a constant speed in a circular orbit of radius R around the earth (mass M). What is the speed of the satellite?

   A. $\sqrt{\frac{GM}{R}}$ C. $\sqrt{\frac{GM}{R^2}}$
   B. $\sqrt{\frac{2GM}{R}}$ D. $\sqrt{\frac{2GM}{R^2}}$

9. 1 pound (as a unit for force) is equal to ____ N.

   A. 2.25 C. 4.45
   B. 3.33 D. 1.45

10. A communications satellite of mass m is orbiting the earth at constant speed in a circular orbit of radius R. If R is increased by a factor of 4, what happens to T, the satellite's orbit period (the time it takes to complete one orbit)?

    A. The period increases by a factor of 2
    B. The period increases by a factor of 4
    C. The period increases by a factor of 8
    D. The period increases by a factor of square root of 8

11. If all of the forces acting on an object balance so that the net force is zero, then

    A. the object must be at rest
    B. the object's speed will decrease
    C. the object's direction of motion can change, but it speed cannot
    D. none of the above

12. If the distance between two point particles is doubled, then the gravitational force between them ___.

    A. decreases by a factor of 4
    B. decreases by a factor of 2
    C. increases by a factor of 2
    D. increases by a factor of 4

13. The planet Pluto has 1/500 the mass and 1/15 the radius of Earth. What is the value of g on the surface of Pluto?

    A. 2.1 $m/s^2$ C. 5.7 $m/s^2$
    B. 4.5 $m/s^2$ D. 9.1 $m/s^2$

14. At the surface of the earth, an object of mass m has weight w. If this object is transported to an altitude that's twice the radius of the earth, then, at the new location,

    A. its mass is m/2 and its weight is w/2
    B. its mass is m/2 and its weight is w/4
    C. its mass is m and its weight is w/4
    D. its mass is m and its weight is w/9

15. You slowly lift a book of mass 2 kg at constant velocity a distance of 3 m. How much work did you do on the book?

    A. 15 J C. 6 J
    B. 30 J D. 60 J

16. A force of 200 N is required to keep an object sliding at a constant speed of 2 m/s across a rough floor. How much power is being expended to maintain this motion?

    A. 100 W C. 800 W
    B. 200 W D. 400 W

17. A ball of mass 2 kg is gently pushed off the edge of a tabletop that is 1.8 m above the floor. Find the speed of the ball as it strikes the floor.

    A. 2 m/s C. 8 m/s
    B. 6 m/s D. 10 m/s

18. A boy (mass = 40 kg) falls off a 50-meter-high cliff. On the way down, the force of air resistance has an average strength of 40 N. Find the speed with which he crashes into the ground.

A. 15 m/s C. 30 m/s
B. 22 m/s D. 44 m/s

19. A pool cue striking a stationary billiard ball (mass = 0.25 kg) gives the ball a speed of 2 m/s. If the force of the cue on the ball was 25 N, over what distance did this force act?

A. 1 cm C. 4 cm
B. 2 cm D. 10 cm

20. A block of mass 3 kg slides down an inclined plane of length 6 m and height 4 m. If the force of friction on the block is a constant 16 N as it slides from rest at the top of the incline, what is its speed at the bottom?

A. 4 m/s C. 12 m/s
B. 6 m/s D. 2 m/s

21. As a rock of mass 4 kg drops from the edge of a 40-meter-high cliff, it experiences air resistance, whose average strength during the descent is 20 N. At what speed will the rock hit the ground?

A. 10 m/s C. 16 m/s
B. 12 m/s D. 20 m/s

22. A force F of strength 20 N acts on an object of mass 3 kg as it moves a distance of 4 m. If F is perpendicular to the 4 m displacement, the work it does is equal to

A. 0 C. 60 J
B. 600 J D. 80 J

23. Two balls roll toward each other. The red ball has a mass of 0.5 kg and a speed of 4 m/s just before impact. The green ball has a mass of 0.3 kg and a speed of 2 m/s. After the head-on collision, the red ball continues forward with a speed of 2 m/s. Find the speed of the green ball after the collision.

A. + 1.3 m/s C. – 1.3 m/s
B. + 2.6 m/s D. – 2.6 m/s

24. Two balls roll toward each other. The red ball has a mass of 0.5 kg and a speed of 4 m/s just before impact. The green ball has a mass of 0.3 kg and a speed of 2 m/s. If the collision is perfectly inelastic, determine the velocity of the composite object after the collision.

A. 1.8 m/s C. 0.8 m/s
B. 2.5 m/s D. 2.9 m/s

25. A block of mass M = 4 kg is moving with velocity V = 6 m/s toward a target block of mass m = 2 kg, which is stationary (v = 0). The objects collide head-on, and immediately after the collision, the speed of block m is 4 times the speed of block M. What is the speed of block M after the collision?

A. 2 m/s C. 3 m/s
B. 1 m/s D. 5 m/s

26. Which of the following best describes a perfectly inelastic collision free of external forces?

A. Total linear momentum is never conserved.
B. Total linear momentum is sometimes (but not always) conserved.
C. Kinetic energy is never conserved.
D. Kinetic energy is sometimes (but not always conserved.

27. Two objects move toward each other, collide, and separate. If there was no net external force acting on the objects, but some kinetic energy was lost, then

A. the collision was elastic and total linear momentum was conserved
B. the collision was elastic and total linear momentum was not conserved
C. the collision was not elastic and total linear momentum was conserved
D. the collision was not elastic and total linear momentum was not conserved

28. A block of mass M = 4 kg is moving with velocity V = 6 m/s toward a target block of mass m = 2 kg, which is stationary (v = 0). The objects collide head-on, and immediately after the collision, the speed of block m is 4 times the speed of block M. What type of collision is this?

A. Elastic
B. Partially Inelastic
C. Totally Inelastic
D. Can not be determined from the available information

29. An 80 kg stuntman jumps out of a window that's 45 m above the ground. He lands on a large, air-filled target, coming to rest in 1.5 s. What average force does he feel while coming to rest?

A. 1.1 kN C. 2.5 kN
B. 1.6 kN D. 3.1 kN

30. Characteristics of simple harmonic motion include which of the following?
I. The acceleration is constant.
II. The restoring force is proportional to the displacement.
III. The frequency is independent of the amplitude.

A. II only C. I and II only
B. I and III only D. II and III only

31. A block of mass m = 1.5 kg is attached to the end of a vertical spring of force constant k = 300 N/m. After the block comes to rest, it is pulled down a distance of 2.0 cm and released. How far does the weight of the block cause the spring to stretch initially?

A. 3 cm C. 6 cm
B. 5 cm D. 10 cm

32. How would the period of the spring–block oscillator change if both the mass of the block and the spring constant were doubled?

A. The period will double
B. The period will be halved
C. The period will quadruple
D. The period will not change

33. A block is attached to a spring and set into oscillatory motion, and its frequency is measured. If this block were removed and replaced by a second block with 1/4 the mass of the first block, how would the frequency of the oscillations compare to that of the first block?

A. f increases by a factor of 2
B. f increases by a factor of 4
C. f increases by a factor of 1.4
D. f decreases by a factor of 1.4

34. A block oscillating on the end of a spring moves from its position of maximum spring stretch to maximum spring compression in 0.25 s. Determine the period and frequency of this motion.

A. 1 Hz C. 3 Hz
B. 2 Hz D. 4 Hz

35. A block attached to an ideal spring undergoes simple harmonic motion. The acceleration of the block has its maximum magnitude at the point where

A. the speed is the maximum
B. the potential energy is the minimum
C. the speed is the minimum
D. the kinetic energy is the maximum

36. How much work would it take to stop an object that has 30 J of kinetic energy?

A. 30 J C. 15 J
B. – 30 J D. – 15 J

37. Momentum is conserved, kinetic energy is NOT conserved, and the objects stick together. This refers to what type of collision.

A. Perfectly Elastic
B. Partially Elastic
C. Partially Inelastic
D. Perfectly Inelastic

38. A person weighs 150 pounds. What is the person's mass?

A. 62 kg C. 68 kg
B. 65 kg D. 72 kg

39. "The ratio of the square of a planet s period of revolution (the time for one complete orbit) to the cube of its average distance from the sun is a constant that is the same for all planets." This statement is known as:

A. Kepler's First Law
B. Kepler's Second Law
C. Kepler's Third Law
D. Kepler's Fourth Law

40. In simple harmonic motion, the frequency is ______ of the amplitude.

A. proportional
B. inversely proportional
C. independent
D. proportional to the square

41. "An object will continue in its state of motion unless compelled to change by a force impressed upon it." This statements is known as:

A. First Law of Motion
B. Second Law of Motion
C. Third Law of Motion
D. The Law of Acceleration

42. A car is traveling in a straight line along a highway. Spotting a police car ahead, the driver of the car slows down from 32 m/s to 20 m/s in 2 seconds. What is the average acceleration of the car?

A. 0 C. - 4 m/s$^2$
B. - 2 m/s$^2$ D. - 6 m/s$^2$

43. An infant crawls 5 m east, then 3 m north, then 1 m east. Find the magnitude of the infant's displacement.

A. 6.0 m C. 6.7 m
B. 8.4 m D. 9.0 m

44. An infant crawls 5 m east, then 3 m north, then 1 m east. If the infant completes his journey in 20 seconds, what is the magnitude of his average velocity?

A. 0.34 m/s C. 0.44 m/s
B. 0.67 m/s D. 0.89 m/s

45. The position on an object (measured in meters from the origin, where x = 0) moving along a straight line is given as a function of time (measured in seconds) by the equation $x(t) = 4t^2 - 6t - 40$. What is the velocity at time t?

A. 0 C. 8t + 6
B. 8t – 6 D. – 8t + 6

46. The position on an object (measured in meters from the origin, where x = 0) moving along a straight line is given as a function of time (measured in seconds) by the equation $x(t) = 4t^2 - 6t - 40$. What is the acceleration at time t?

A. 0 C. 6 m/s$^2$
B. 8 m/s$^2$ D. 8t m/s$^2$

47. What is the upward acceleration of an elevator in which a 90 kg man inside will cause a pressure of 1000 N on the its floor?

A. 0.7 m/s2 C. 1.1 m/s2
B. 0.9 m/s$^2$ D. 1.3 m/s$^2$

48. A car (mass = 1800 kg) is traveling at a speed of 60 kph. What force is necessary to reduce its acceleration at a rate of 20 cm/s$^2$?

A. 240 N C. – 240 N
B. 360 N D. – 360 N

49. Two blocks of mass 6 kg and m kg, are placed in a location where g = 9.67 m/s$^2$. The combined weight of the two blocks is 174 N. Find the value of m.

A. 12 kg C. 16 kg
B. 14 kg D. 18 kg

50. What is the combined kinetic energy of a motorcycle with mass of 100 kg and with two passengers with a combined mass of 125 kg if it is moving at a speed of 40 kph?
A. 12 kJ C. 22 kJ
B. 14 kJ D. 24 kJ